INTERNAL REPORT: PRELIMINARY SAFETY, PHARMACOKINETICS, AND TOXICITY EVALUATION OF AMYLOID-TARGETING COMPOUND CPH-412 (ALZHEIMER'S THERAPEUTIC PIPELINE - PHASE 0.1)

Document ID: CYM-PH-2026-CPH412-TOX-01
Classification: STRICTLY CONFIDENTIAL - INTERNAL R&D ONLY
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Date: May 22, 2026

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1. Executive Summary

Compound CPH-412 is a novel, small-molecule, blood-brain barrier (BBB) penetrating modulator of amyloid-beta (Aβ) aggregation, designed for the disease-modifying treatment of early-onset and prodromal Alzheimer's Disease. This preliminary phase 0.1 toxicity evaluation compiles results from in vitro binding affinity assays, in vivo rodent and non-human primate (NHP) safety margins, and early human microdosing cohorts (Phase 0.1).

The primary objective of this report is to evaluate the safety, tolerability, and pharmacokinetic (PK) profiles of CPH-412 across escalating dose bands to guide Phase 1 clinical trial designs.

Overall, CPH-412 demonstrated excellent central nervous system (CNS) penetration and robust target engagement. However, key safety flags arose in the highest-exposure human cohort (designated as **Group A14**). Notably, 15% of subjects in this cohort exhibited marked, transient elevations in hepatic transaminases—specifically alanine aminotransferase (ALT) and aspartate aminotransferase (AST)—exceeding 3 times the upper limit of normal (ULN). This report analyzes these findings, discusses the suspected hepatic metabolic saturation pathways, and provides actionable dosage adjustments and monitoring recommendations for subsequent clinical protocols.

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2. Target Mechanism & Chemical Profile

CPH-412 [chemical IUPAC name: 2-(4-(3-(trifluoromethyl)phenyl)piperazin-1-yl)pyrimidin-5-yl methylcarbamate] operates as a highly selective, non-competitive antagonist of soluble Aβ oligomerization.

None
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          [CF3-Phenyl] - [Piperazine] - [Pyrimidine] - [Carbamate Linker]
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2.1 Receptor Binding and Selectivity

In vitro radioligand binding studies indicate that CPH-412 exhibits nanomolar affinity ($K_i = 4.2 \text{ nM}$) for soluble $A\beta_{42}$ assemblies, with $>1000\text{-fold}$ selectivity over other physiological fibrillar aggregates (such as tau, alpha-synuclein, and huntingtin). CPH-412 does not exhibit significant binding to major cardiac ion channels ($hERG$ $IC_{50} > 50\ \mu\text{M}$), suggesting a low risk of drug-induced QT prolongation.

2.2 Mechanism of Action

By binding to the hydrophobic cleft of early-stage Aβ monomeric clusters, CPH-412 sterically inhibits the nucleation-dependent polymerization process. This blocks the cascade that leads to neurotoxic protofibril formation. In cortical neuron cultures, pre-treatment with CPH-412 at 50 nM completely rescued synaptic plasticity deficits induced by synthetic $A\beta_{42}$ oligomer exposure.

2.3 Drug-Drug Interaction Risk

Given the compound's high reliance on CYP3A4 for primary metabolism, a high potential for clinically significant drug-drug interactions (DDIs) is noted, particularly with strong CYP3A4 inhibitors (e.g., ketoconazole, clarithromycin). Pre-screening for concomitant medications and a dedicated DDI study are critical for Phase 1b planning.

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3. Pre-Clinical In Vitro and In Vivo Profiling

Before initiating human microdosing, comprehensive pre-clinical toxicology studies were conducted in Sprague-Dawley rats and Cynomolgus monkeys.

3.1 Rodent ADME and Toxicity

Rats were administered oral daily doses of 10, 50, and 200 mg/kg of CPH-412 for 28 consecutive days.

- **Absorption:** Rapid absorption was observed with $T_{\max}$ occurring at 1.5 hours post-dose. Oral bioavailability (F) was determined to be approximately $68%$.
- **Distribution:** Volume of distribution (V_d) was high (3.4 L/kg), indicating extensive tissue distribution. Brain-to-plasma ratios reached $1.8 : 1.0$ at steady state, confirming highly efficient BBB penetration.
- **Excretion:** The primary route of excretion was fecal (75%), suggesting significant biliary clearance, with 25% cleared renally.
- **Toxicity Findings:** No mortality was recorded in the 10 and 50 mg/kg groups. In the 200 mg/kg high-dose group, mild-to-moderate follicular hypertrophy of the thyroid gland was noted, which resolved following a 14-day recovery period. No evidence of hepatotoxicity or histopathological hepatic lesions was detected in rodent models at any dose level.

3.2 Non-Human Primate (NHP) Safety Margins

Cynomolgus monkeys received escalating intravenous and oral doses ranging from 5 mg/kg to 80 mg/kg over a 14-day period.

- **Cardiovascular Safety:** Telemetric monitoring showed no clinically significant changes in mean arterial pressure, heart rate, or ECG intervals (PR, QRS, QTc).
- **Neurological Tolerance:** No abnormal behavior, tremors, or ataxia were noted up to the highest dose (80 mg/kg).
- **No Observed Adverse Effect Level (NOAEL):** The pre-clinical NOAEL in NHPs was established at 40 mg/kg/day, providing a safety margin of approximately 35-fold over the anticipated initial therapeutic human exposure.

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4. Phase 0.1 Clinical Protocol and Patient Demographics

A randomized, double-blind, placebo-controlled, single-ascending-dose (SAD) study was conducted in healthy adult volunteers (ages 18 to 45) to evaluate the safety, tolerability, and pharmacokinetics of CPH-412.

4.1 Cohort Stratification

A total of 100 subjects were randomized across five escalating dose cohorts (Groups A10 through A14), receiving either active CPH-412 or matched placebo in an 8:2 ratio.

Cohort Group	Number of Active Subjects	Number of Placebo Subjects	Dose Level administered (Single Dose)	Mode of Administration
Group A10	16	4	25 mg	Oral capsule (fasted)
Group A11	16	4	50 mg	Oral capsule (fasted)
Group A12	16	4	100 mg	Oral capsule (fasted)
Group A13	16	4	200 mg	Oral capsule (fasted)
Group A14	16	4	400 mg	Oral capsule (fasted)

4.2 Subject Demographics

Exclusion criteria included any pre-existing hepatic, renal, or cardiovascular disease, history of alcohol abuse, or concurrent use of medications metabolized extensively by cytochrome P450 enzymes. The cohorts were balanced for age, sex, and body mass index (BMI).

5. Pharmacokinetic (PK) and Pharmacodynamic Analysis

Pharmacokinetic sampling was performed at pre-dose, and at 0.5, 1, 2, 4, 6, 8, 12, 24, 48, and 72 hours post-dose.

5.1 Linear and Non-Linear Dose Proportionality

Up to the 200 mg dose level (Groups A10 to A13), CPH-412 exhibited highly linear and dose-proportional pharmacokinetics. Double the dose resulted in a precise doubling of the Maximum Plasma Concentration ($C_{\max}$) and the Area Under the Curve ($AUC_{0-\infty}$).

However, in the **Group A14** cohort (400 mg dose), a non-linear pharmacokinetic shift was observed:

- $C_{\max}$ increased from $1.1\ \mu\text{g/mL}$ (in Group A13) to a disproportionate $3.8\ \mu\text{g/mL}$ (in Group A14).
- $AUC_{0-\infty}$ increased by approximately 4.2-fold compared to the 200 mg group, indicating a reduction in clearance (Cl) at the 400 mg dose limit.
- Plasma half-life ($t_{1/2}$) widened from 14.5 hours (Group A13 average) to 26.8 hours (Group A14 average).

This abrupt decrease in drug clearance strongly suggests the saturation of primary hepatic metabolic clearance pathways at systemic concentrations exceeding $2.5\ \mu\text{g/mL}$.

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6. Toxicity and Safety Evaluation

Standard clinical safety monitoring—including vital signs, physical examinations, hematology, urinalysis, and comprehensive metabolic panels—was performed daily for 7 days post-dose.

6.1 Adverse Event (AE) Frequency Summary

CPH-412 was generally well-tolerated at dose levels up to 200 mg. The most common mild adverse events across all groups were mild headache (8%), nausea (4%), and localized abdominal discomfort (3%). These events resolved spontaneously and did not require medical intervention.

6.2 Group A14 Safety Profile & Anomalies

A significant safety signal emerged exclusively within **Group A14 (the 400 mg dosing cohort)**. Of the 16 subjects who received the active 400 mg dose, three subjects (approximately 15% of the total 20-subject cohort) developed clinically significant elevations in liver transaminases.

These three subjects (Subject ID #1404, #1409, and #1411) exhibited the following clinical milestones:

- **Onset:** Transaminase elevations began at Day 3 post-dose, peaking on Day 5, and returning to baseline by Day 12.
- **Symptomatology:** All three subjects remained completely asymptomatic. They reported no abdominal pain, fatigue, fever, pruritus, or dark urine.
- **Other Markers:** Serum bilirubin, alkaline phosphatase (ALP), and gamma-glutamyl transferase (GGT) remained within normal physiological limits for all subjects. International Normalized Ratio (INR) was stable, ruling out acute hepatic functional impairment.

Because transaminase elevation occurred in isolation without concurrent elevations in bilirubin ($>2 \times \text{ULN}$), these cases do not meet the criteria for **Hy's Law**, which would indicate a high risk of drug-induced severe liver injury. However, the 15% incidence rate represents a crucial safety threshold that must be addressed before proceeding to Phase 1b/2a multi-dose studies.

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7. Hepatotoxicity and Liver Enzyme Fluctuations

To understand the biological basis of the safety signals observed in Group A14, a detailed comparative biochemical review was conducted.

7.1 Quantitative Analysis of Enzyme Fluctuations

The table below tracks the temporal progression of AST and ALT elevations (expressed as multiples of the Upper Limit of Normal) for the three affected subjects in Group A14 compared to the cohort average.

Day Post-Dose	Subject 1404 (ALT / AST)	Subject 1409 (ALT / AST)	Subject 1411 (ALT / AST)	Cohort A14 Mean (Excluding Placebo)
Pre-dose	0.8x / 0.9x	0.7x / 0.6x	0.9x / 0.8x	0.8x / 0.7x
Day 1	1.1x / 1.0x	1.0x / 0.9x	1.2x / 1.1x	1.0x / 0.9x
Day 3	2.4x / 2.1x	1.9x / 1.8x	2.8x / 2.5x	1.4x / 1.2x
Day 5 (Peak)	**4.2x / 3.8x**	**3.6x / 3.2x**	**4.9x / 4.1x**	**1.9x / 1.6x**
Day 7	3.1x / 2.8x	2.5x / 2.2x	3.5x / 2.9x	1.5x / 1.3x
Day 10	1.6x / 1.4x	1.2x / 1.1x	1.8x / 1.5x	1.1x / 1.0x
Day 14	0.9x / 0.9x	0.8x / 0.7x	1.0x / 0.9x	0.8x / 0.8x

7.2 Suspected Cellular Mechanism of Hepatotoxicity

CPH-412 is primarily metabolized by the cytochrome P450 3A4 (**CYP3A4**) isoform, yielding an inactive carboxylated metabolite.

We hypothesize that at the single high dose of 400 mg (administered to Group A14), the rate of drug entry into hepatic tissue exceeds the enzymatic capacity of CYP3A4. This metabolic bottleneck triggers several cellular pathways:

1. **Enzymatic Saturation:** As CYP3A4 saturates, CPH-412 is forced into alternative, minor metabolic pathways (such as CYP2D6 and CYP2C19).
2. **Oxidative Stress:** These secondary pathways generate high levels of reactive oxygen species (ROS) within hepatocytes, causing transient oxidative stress.
3. **Membrane Leakage:** This localized ROS stress compromises the permeability of hepatocyte cell membranes. This causes AST and ALT enzymes to leak into the systemic circulation without causing cell death (hepatocyte necrosis). This transient leakage explains the rapid decline in enzyme levels as the systemic drug concentration falls below the saturation threshold of $2.5\ \mu\text{g/mL}$.

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8. Recommendations and Dose-Adjustment Protocols

To maintain the efficacy of CPH-412 against Aβ aggregation while eliminating the risk of hepatotoxicity, subsequent clinical phases must prevent systemic concentrations from crossing the **$2.5\ \mu\text{g/mL}$ saturation threshold**.

8.1 Split-Dose Regimen

Rather than administering a single high dose of 400 mg, future protocols should evaluate a **split-dosing regimen of 150 mg twice daily (BID)**, totaling 300 mg per day.

- **PK Modeling:** Simulating a 150 mg BID regimen predicts a steady-state $C_{\max}$ of approximately $1.6\ \mu\text{g/mL}$. This remains safely below the CYP3A4 saturation threshold while keeping the trough concentration ($C_{\min}$) above the 50 nM minimum required for full therapeutic target engagement in the brain.

[Figure 8.1: Simulated Steady State PK Profile. This figure illustrates predicted plasma concentration (Cmax) over time for the proposed 150 mg BID regimen (reaching a peak of 1.6 ug/mL) compared to the single 400 mg QD dose (peak of 3.8 ug/mL). The 2.5 ug/mL saturation threshold is marked.]

8.2 Intermittent Dosing Schedule

Alternatively, an **intermittent schedule of 5 days on, 2 days off** using a 200 mg daily dose should be explored. This 2-day wash-out period allows hepatic glutathione reserves to fully replenish, preventing the accumulation of reactive oxidative metabolites.

8.3 Inclusion of Co-Factors

For high-exposure cohorts, concurrent administration of 600 mg of N-acetylcysteine (NAC) daily may be considered. NAC is a direct precursor to glutathione. It acts as an antioxidant that can neutralize the ROS generated during secondary CYP metabolism, neutralizing membrane leakage.

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9. References & Appendices

1. Vance, E. et al. (2025). *In Vitro Binding Characteristics of CPH-412 to Amyloid-Beta Oligomers.* Cymbal Pharma Internal R&D Archive.
2. *Non-Clinical Toxicology Report: 28-Day Oral Study of CPH-412 in Sprague-Dawley Rats.* (Document ID: CYM-TOX-RAT-412).
3. *Non-Clinical Toxicology Report: 14-Day Escalating Dose Study of CPH-412 in Cynomolgus Monkeys.* (Document ID: CYM-TOX-NHP-412).
4. FDA Guidance for Industry: *Drug-Induced Liver Injury: Premarketing Clinical Evaluation.* (July 2009).